

A Conditional Deficit-Summand Reduction For The Skew-Hermitian Dimension Problem

FA/Banach discovery run `fa_banach_001`

Conditional packet, 2026-06-13

Obsolescence Warning

This appendix is retained as development provenance for the all-dimensional Problem 27 route. It is not a current proof of Problem 27. The deficit-summand formulation used below is too strong: it asks for an independent proper invariant summand. This fails, for example, for $U(2) \leq SO(4)$ on $\mathbb{C}_{\mathbb{R}}^2$, where

$$\dim U(2) = 4, \quad w(4) = 5 > 4,$$

but the real representation is irreducible. The corrected route, recorded in the root packet, must allow $W = V$, must not require W to be G -invariant, and must include a strict enlargement clause. The centralizer computations below remain useful evidence, but the theorem below should not be cited as a completed proof.

Source problem

The source paper is V. Kadets, M. Martin, and R. Paya, “Recent progress and open questions on the numerical index of Banach spaces”, arXiv:math/0605781. Problem 27 asks for the possible values of $\dim \mathcal{Z}(X)$ for finite-dimensional real Banach spaces.

Our next aim is to discuss some questions related to the Lie algebra of skew-hermitian operators $\mathcal{Z}(X)$ of an arbitrary n -dimensional space which. The main related open question is the following.

Problem 27. *Figure out what are the possible values for the dimension of $\mathcal{Z}(X)$ when $\dim(X) = n$.*

Let us fix a n -dimensional Banach space X . It follows from a theorem of Auerbach [72, Theorem 9.5.1], that there exists an inner product $(\cdot|\cdot)$ on X such that every skew-hermitian operator on X remains skew-hermitian (hence skew-symmetric) on $H := (X, (\cdot|\cdot))$. Then, by just fixing an orthonormal basis of H , we get an identification of $\mathcal{Z}(X)$ with a Lie subalgebra of the Lie algebra $\mathcal{A}(n)$. Therefore,

$$\dim(\mathcal{Z}(X)) \leq \frac{n(n-1)}{2}.$$

The equality holds if and only if X is a Hilbert space (see [73, Theorem 3.2] or [60, Corollary 2.7]). It is a good question whether or not all the intermediate numbers are possible values for the dimension of $\mathcal{Z}(X)$. The answer is negative, as a consequence of Theorem 3.2 in Rosenthal's paper [73], which reads as follows.

- (a) If $\dim(\mathcal{Z}(X)) > \frac{(n-1)(n-2)}{2}$, then X is a Hilbert space and so $\dim(\mathcal{Z}(X)) = \frac{n(n-1)}{2}$.
- (b) $\dim(\mathcal{Z}(X)) = \frac{(n-1)(n-2)}{2}$ if and only if X is a non-Euclidean absolute sum of $\mathbb{R}$ and a Hilbert space of dimension $n-1$.

For low dimensions, Problem 27 has been solved in [74]. When the dimension of X is 3, the above result leaves only the following possible values for the dimension of $\mathcal{Z}(X)$: 0 as for $X = \ell_\infty^3$, 1 as for $\mathbb{R} \oplus_1 \mathbb{C}$, and 3 as for ℓ_2^3 . When the dimension of X is 4, the possible values of the dimension of $\mathcal{Z}(X)$ allowed by the above result are 0, 1, 2, 3, 6; all of them are possible [74, pp. 443]. The first dimension in which Problem 27 is open is $n = 5$.

Problem 28. *What are the possible values for the dimension of $\mathcal{Z}(X)$ when X is a 5-dimensional real Banach space?*

Triangular support notation

Let

$$T_k = \binom{k}{2} \quad (k \geq 2)$$

and define

$$w(m) = \min \left\{ \sum_j k_j : m = \sum_j \binom{k_j}{2}, k_j \geq 2 \right\},$$

with $w(0) = 0$. Then m belongs to

$$P(n) = \left\{ \sum_j \binom{k_j}{2} : \sum_j k_j = n \right\}$$

if and only if $w(m) \leq n$.

The expected answer to Problem 27 is:

$$\dim \mathcal{Z}(X) \in P(n) \quad \text{for every } n\text{-dimensional real Banach space } X,$$

and every value in $P(n)$ is realized by an ℓ_∞ -sum of Euclidean blocks.

Remaining dependency and proved coupling lemma

Conditional Dependency 1 (Small-Module Lemma). *Let $H \leq \mathrm{SO}(E)$ be compact, connected, effective, and irreducible over the real numbers. If*

$$w(\dim H) > \dim E,$$

then, up to harmless full orthogonal blocks and low-rank isomorphisms, the action is one of the standard dangerous sphere-transitive atoms:

$$\mathrm{SU}(k) \text{ or } \mathrm{U}(k) \curvearrowright \mathbb{C}^k, \quad \mathrm{U}(2) = \mathrm{Sp}(1) \mathrm{U}(1) \curvearrowright \mathbb{H}, \quad G_2 \curvearrowright \mathbb{R}^7.$$

Lemma 1 (Centralizer Coupling Lemma). *Let A be one of the dangerous standard atoms in Assumption 1, acting on a real block E . Put*

$$a = \dim A, \quad e = \dim E.$$

Let S be the transitive semisimple factor of A . Suppose S acts nontrivially on a second irreducible real summand

$$F = U \otimes_D D^m,$$

where U is an irreducible real S -module and $D = \mathrm{End}_S(U) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$. Let

$$C_D(m) = \begin{cases} \dim \mathrm{SO}(m) = \binom{m}{2}, & D = \mathbb{R}, \\ \dim \mathrm{U}(m) = m^2, & D = \mathbb{C}, \\ \dim \mathrm{Sp}(m) = m(2m+1), & D = \mathbb{H}. \end{cases}$$

Then

$$\max_{0 \leq \ell \leq C_D(m)} w(a + \ell) \leq e + m \dim_{\mathbb{R}} U.$$

Proof. We use two elementary triangular-weight bounds. First, for every $p \geq 3$ and every $0 \leq N \leq \binom{p}{2} + 4$,

$$w(N) \leq 2p + 2. \tag{1}$$

To prove this, note that $w(N) \leq N$ for every $N \geq 6$: check $N = 6, 7, 8$ directly, and then subtract $\binom{3}{2} = 3$ inductively. Now prove (1) by induction on p . The case $p = 3$ is direct, since the maximum on $0 \leq N \leq 7$ is $w(5) = 7 \leq 8$. If $p \geq 4$ and $N \leq \binom{p-1}{2} + 4$, use induction. Otherwise write $N = \binom{p-1}{2} + q$, where $5 \leq q \leq p + 3$. If $q = 5$, then $w(q) = 7 \leq p + 3$; if $q \geq 6$, then $w(q) \leq q \leq p + 3$. Hence $w(N) \leq (p-1) + w(q) \leq 2p + 2$.

Second, for every $m \geq 1$ and every $0 \leq N \leq m^2$,

$$w(N) \leq 4m. \tag{2}$$

The cases $m = 1, 2, 3$ are direct. For $m \geq 4$, if $N \leq \binom{m}{2}$, (1) gives $w(N) \leq 2m + 2 \leq 4m$. If $N = \binom{m}{2} + q$, then $q \leq m^2 - \binom{m}{2} = \binom{m+1}{2}$, so (1) gives $w(q) \leq 2m + 4$ and hence $w(N) \leq m + 2m + 4 \leq 4m$.

We now check the dangerous atoms. For the $\mathrm{U}(2)$ atom, the smallest real-type occurrence of the semisimple factor $\mathrm{SU}(2)$ has real dimension 3, and the smallest quaternionic-type occurrence has real dimension 4. In the real centralizer case, $a = 4$, $e = 4$, and $C_{\mathbb{R}}(m) = \binom{m}{2}$. For $m = 1, 2$ the inequality is direct; for $m \geq 3$, (1) gives

$$w(4 + \ell) \leq 2m + 2 \leq 4 + 3m.$$

In the quaternionic case, $C_{\mathbb{H}}(m) = m(2m + 1) = \binom{2m+1}{2}$, and (1) with $p = 2m + 1$ gives

$$w(4 + \ell) \leq 4m + 4.$$

For the G_2 atom, $a = 14$, $e = 7$, the smallest occurrence has real dimension 7, and $C_{\mathbb{R}}(m) = \binom{m}{2}$. The case $m = 1$ is $w(14) = 10 \leq 14$. For $m \geq 2$,

$$14 + \ell \leq 14 + \binom{m}{2} \leq \binom{m+6}{2} + 4,$$

so (1) gives $w(14 + \ell) \leq 2m + 14 \leq 7m + 7$.

For $SU(k)$, $k \geq 3$, with its standard coupled occurrence, we have $a = k^2 - 1$, $e = 2k$, $\dim_{\mathbb{R}} U = 2k$, and $C_{\mathbb{C}}(m) = m^2$. The identity

$$k^2 - 1 = \binom{k+1}{2} + \binom{k-1}{2} + (k-2)\binom{2}{2}$$

gives $w(k^2 - 1) \leq 4k - 4$. By (2),

$$w(k^2 - 1 + \ell) \leq 4k - 4 + 4m \leq 2k + 2km,$$

because the difference is $2(k-2)(m-1) \geq 0$.

For $U(k)$, $k \geq 3$, the identity $k^2 = \binom{k+1}{2} + \binom{k}{2}$ gives $w(k^2) \leq 2k + 1$, and (2) gives

$$w(k^2 + \ell) \leq 2k + 1 + 4m \leq 2k + 2km.$$

Finally, larger coupled irreducible modules only increase the right side for their centralizer type. The standard minimal-module facts used here are: the $SU(2)$ real and quaternionic minima are 3 and 4; the G_2 minimum is 7 and is real type; and for $SU(k)$ the standard complex module is the smallest complex-type occurrence, while nonstandard real or quaternionic occurrences are large enough to make the displayed estimates easier. The only smaller real-type low-rank exception is $SU(4)$ on $\Lambda^2 \mathbb{C}^4$, but the corresponding standard $SU(4)$ and $U(4)$ atoms are not dangerous, since $w(15) = 6 \leq 8$ and $w(16) = 8$. $\square$

Conditional theorem

Theorem 1. *Assume 1. Then for every n -dimensional real Banach space X ,*

$$\dim \mathcal{Z}(X) \in P(n).$$

Consequently, under this assumption, the possible values of $\dim \mathcal{Z}(X)$ are exactly the elements of $P(n)$.

Proof. The realization direction is unconditional. If $n = \sum_j k_j$, take the ℓ_∞ -sum of Euclidean spaces of dimensions k_j . Its connected isometry group is the product of the orthogonal groups on the Hilbert blocks, so the Lie algebra dimension is $\sum_j \binom{k_j}{2}$.

It remains to prove the obstruction. Let $G = \text{Iso}_0(X)$ and choose an Auerbach inner product so that $G \leq \text{SO}(V)$, where V is the underlying real vector space of X . It is enough to prove the following deficit-summand theorem:

If $G \leq \text{SO}(V)$ is compact, connected, effective, and $w(\dim G) > \dim V$, then V has an orthogonal summand W and an ideal of $\text{Lie}(G)$ that acts trivially on $W^\perp$ and acts on W through a proper compact connected sphere-transitive group.

Indeed, for each fixed $u \in W^\perp$, the function $x \mapsto \|u + x\|$ on W is constant on Euclidean spheres. Hence the norm is invariant under the full $\mathrm{SO}(W)$ on that summand. Since the projected group was proper sphere-transitive, this enlarges the connected isometry group, a contradiction. Therefore $w(\dim G) \leq \dim V$.

We now prove the deficit-summand theorem. Suppose it fails, and choose a counterexample $G \leq \mathrm{SO}(V)$ with the number of irreducible real G -summands in V minimal. Write

$$V = E_1 \oplus \cdots \oplus E_r.$$

There is no independent proper sphere-transitive summand by assumption.

Let H_i be the image of G on E_i , and let K_i be the kernel of this projection. If $w(\dim H_i) \leq \dim E_i$, then K_i acts on $E_i^\perp$. Any independent sphere-transitive summand for this action would also be independent for G , because K_i fixes E_i . By minimality,

$$w(\dim K_i) \leq \dim E_i^\perp.$$

Subadditivity of w gives

$$w(\dim G) \leq w(\dim H_i) + w(\dim K_i) \leq \dim E_i + \dim E_i^\perp = \dim V,$$

contradiction. Hence every image H_i is dangerous. By Assumption 1, every H_i is a standard dangerous sphere-transitive atom.

Fix one summand $E = E_i$, let $A = H_i$, and let S be the transitive semisimple factor of A . If the lifted ideal S acts trivially on $E^\perp$, then E is an independent proper sphere-transitive summand, contrary to our choice of counterexample. Thus S acts nontrivially on some other irreducible summand F .

Let Q be the image of G on $E \oplus F$. On E it contains the atom A . On F , the same semisimple factor S acts through a nontrivial irreducible module U , and the remaining commuting image lies in the full centralizer $\mathrm{SO}(m)$, $\mathrm{U}(m)$, or $\mathrm{Sp}(m)$ according to the division algebra $D = \mathrm{End}_S(U)$. Hence

$$\dim Q \leq a + \ell$$

for some $0 \leq \ell \leq C_D(m)$. Lemma 1 gives

$$w(\dim Q) \leq w(a + \ell) \leq \dim E + \dim F.$$

Let N be the kernel of the projection of G to $E \oplus F$. Again, any independent sphere-transitive summand for N on $(E \oplus F)^\perp$ would be independent for G . Minimality gives

$$w(\dim N) \leq \dim(E \oplus F)^\perp.$$

Therefore

$$w(\dim G) \leq w(\dim Q) + w(\dim N) \leq \dim E + \dim F + \dim(E \oplus F)^\perp = \dim V,$$

the final contradiction. This proves the deficit-summand theorem and hence the obstruction direction. $\square$

Evidence and verification

The Small-Module Lemma is standard representation theory. The supporting attempt note reduces it to the usual small-representation tables and gives explicit estimates such as

$$w(k^2 - 1) \leq 4k - 4, \quad w(k^2) \leq 2k + 1, \quad w(\dim \mathrm{Sp}(k)) \leq 2k + 1.$$

The script `code/weyl_small_irrep_scan.py` checks the classical highest-weight statement through rank 80.

The Centralizer Coupling Lemma is proved above. One script checks the worst possible integer ℓ up to the full centralizer dimension for dangerous atoms with $k \leq 300$ and $m \leq 300$. A second script checks the two triangular-weight bounds in Lemma 1 through $p, m \leq 500$ and verifies the real/quaternionic centralizer families over the same range. The original stress test reports no failures:

Centralizer-coupling families checked: 3900, Failures: 0.

These scripts are verification aids; the proof of Lemma 1 is the written interval argument above.

Verification notes

The proof of Theorem 1 is formal after the remaining Small-Module Lemma. Human review should focus on:

1. proving or citing Assumption 1;
2. checking the triangular-number interval proof of Lemma 1;
3. checking that the minimal-counterexample induction correctly preserves the independence condition when passing to kernels.

Novelty and limitation

This packet is not an unconditional solution of Problem 27. It is a conditional reduction with one explicitly stated unproved dependency, the Small-Module Lemma. Local run-index searches for 0605781, `dim Z(X)`, Problem 27, `deficit`, and `skew-hermitian` found only the existing five-, six-, and seven-dimensional packets and the attempt notes used here. Web/literature searches during the attempt found low-degree representation sources such as Guralnick–Larsen–Manack, but no later paper explicitly resolving Problem 27.